Jayesh M. K.

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Summary

Speech & Audio AI Engineer with a PhD in Electrical Engineering and extensive experience developing and commercializing real-time speech and audio AI systems. Expertise in ASR, speech enhancement, source separation, VAD, endpoint detection, voice cloning, beamforming, and AEC, with deep-learning optimization for edge deployment. Experienced in leading engineering teams and driving AI innovations from research to production.

Work Experience

Speech Engineer — Omilia, Independent Contractor, Remote Oct 2025 - Aug 2026

- Fine-tuned the Parakeet ASR model to improve noise robustness for drive-thru application, improving WER in challenging acoustic environments.
- Collaborated with the platform team to implement a word-boosting strategy, delivering a 12% relative improvement in order-pickup-related escalations by addressing name recognition errors.
- Implemented acoustic-only turn-detection module, eliminating reliance on linguistic features to enable fast, low-latency endpointing.

Staff Engineer — Samsung R&D Institute India-Bangalore Jan 2019 - Sep 2025

- Led and mentored a team of up to four engineers, overseeing product development and delivery from 2023–2025.
- Led the development of a deep-learning-based spatial speech separation model for VR glasses, driving the end-to-end pipeline from speech data collection and curation through model development, training, and Android optimization, achieving an on-device inference latency of < 0.05 s.
- Led the development and commercialization of a low-latency, LSTM-based noise-robust streaming VAD, deployed across multiple Bixby-enabled Samsung devices.
- Personalized endpoint detector. Model voice activity of a target speaker at the frame level.
- Developed and commercialized a Speech Endpoint Detector for voice-assistant systems, integrating VAD and Partial ASR output
- Worked on improving neural-network-based multi channel speech enhancement and optimized for stringent model-size and CPU-utilization constraints.
- Developed a DNN-based Acoustic Echo Cancellation (AEC) system for robust barge-in handling, capable of operating under extremely low signal-to-echo ratios down to -30 dB.
- Developed low-latency embedded inference pipelines for deep learning based speech front-ends (VAD, AEC, noise suppression, and beamforming) on Tizen OS, utilizing the TFLite `C++` runtime and hardware acceleration.

  Leadership
  - Mentored engineers and coordinated research, development, and productization activities.
  - Spearheaded collaborations with academic institutes IISC and IIT Kanpur that resulted in research publications and technology contributions to products.

Education

2010 – 2018 PhD in Electrical Engineering, **IIT Madras**
Thesis: New Methods for Vocal-tract Analysis
Research focused on vocal-tract estimation and new features for ASR.

2005 – 2009 B.Tech in Electronics and Communication, **Govt. College of Engineering, Kannur**

Publications

Journals

- Shaahid Ahmed Nadeem, Nithya Darisini Paruvachi Subramanian, Mohanaprasad Kothandaraman, M. K. Jayesh, et al. (2026). "Multi-Reference Echo Cancellation". In: *IEEE Access* 14, pp. 103450–103461. DOI: 10.1109/ACCESS.2026.3709458
- M. K. Jayesh and C. S. Ramalingam (Aug. 2017). "A One-Dimensional Search Method with Stable 1-Norm Solution for Linear Prediction". In: *The Journal of the Acoustical Society of America* 142.2, EL170–EL176. DOI: 10.1121/1.4996455
- M. K. Jayesh and C. S. Ramalingam (July 2016). "Improved Chirp Group Delay Based Algorithms with Applications to Vocal Tract Estimation and Speech Recognition". In: *Speech Communication* 81, pp. 72–89. DOI: 10.1016/j.specom.2016.02.003

Conferences

- Prithvi R. R. Gudepu, Jayesh M. Koroth, Kamini Sabu, and Mahaboob Ali Basha Shaik (2023). "Dynamic Encoder RNN for Online Voice Activity Detection in Adverse Noise Conditions". In: *Interspeech 2023*, pp. 5052–5056. DOI: 10.21437/Interspeech.2023-2466
- M. K. Jayesh, Mukesh Sharma, Praneeth Vonteddu, Mahaboob Ali Basha Shaik, et al. (2022). "Transformer Networks for Non-Intrusive Speech Quality Prediction". In: *Interspeech 2022*, pp. 4078–4082. DOI: 10.21437/Interspeech.2022-10020
- Mudit D. Batra, M. K. Jayesh, and C. S. Ramalingam (2022). "Robust Pitch Estimation Using Multi-Branch CNN-LSTM and 1-Norm LP Residual". In: *Interspeech 2022*, pp. 3573–3577. DOI: 10.21437/Interspeech.2022-10704
- R. G. Prithvi Raj, Rohit Kumar, M. K. Jayesh, Anurenjan Purushothaman, et al. (2021). "SRIB-LEAP Submission to Far-Field Multi-Channel Speech Enhancement Challenge for Video Conferencing". In: *Interspeech 2021*, pp. 1857–1861. DOI: 10.21437/Interspeech.2021-1283
- M. K. Jayesh and C. S. Ramalingam (2014). "An Improved Chirp Group Delay Based Algorithm for Estimating the Vocal Tract Response". In: *2014 Proceedings of the 22nd European Signal Processing Conference (EUSIPCO)*. Lisbon, Portugal, pp. 2295–2299

Patents

- Midhun Sreekumar Menon, Viswanath Veera , Jayesh Mundayadan Koroth, Rajath C Aralikatti, Srinidhi N , Lakshmi Priya Muraleedharan "Intrinsic3D: A method for providing live user guidance for data capture for joint spatial-acoustic-illumination XR scene reconstruction" (US patent filed, 2026)

Skills

Programming	Python, C++, MATLAB, Bash
Machine Learning	PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, SciPy
Speech & Audio AI	ASR, TTS, Speech Enhancement, Noise Suppression, AEC, Beamforming, Speech Separation, VAD, Speaker Diarization, Audio Classification, Voice Cloning
Speech Toolkits	Kaldi, SpeechBrain, NVIDIA NeMo, torchaudio
Optimization & Deployment	ONNX, TensorRT, Quantization, Knowledge Distillation, Model Pruning, GPU/CPU Optimization, Edge AI, Real-time Inference
Software Engineering	Git, GitLab, Linux, VS code, cursor, Android studio
Cloud, MLOps & Data	AWS, Snowflake, Kubeflow, ML Pipelines
Signal Processing	Probability, Linear Algebra, Digital Signal Processing, Speech Signal Processing, Convex Optimization, Estimation Theory, Adaptive Signal Processing
AI-Assisted Development	Cursor, Claude

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